

Chapter 1: Neural Networks

A neural network is a computational model inspired by the human brain. It consists of layers of interconnected nodes, or neurons. Each connection has a weight that adjusts as the network learns from data. The process of adjusting these weights is called training, and it typically uses an algorithm called backpropagation combined with gradient descent.

Chapter 1 continued

Neural networks are organized into an input layer, one or more hidden layers, and an output layer. The input layer receives raw data. Hidden layers transform this data through weighted sums and activation functions. Common activation functions include ReLU, sigmoid, and tanh. The output layer produces the final prediction, whether that is a classification label or a continuous value.

Chapter 2: Retrieval-Augmented Generation

Retrieval-Augmented Generation, or RAG, combines a retrieval system with a language model. Instead of relying purely on parameters memorized during training, the model retrieves relevant documents at inference time and conditions its output on that retrieved context. This reduces hallucination and allows the system to cite specific sources for its claims.